

**BEFORE THE NATIONAL GREEN TRIBUNAL
PRINCIPAL BENCH, NEW DELHI**

(HYBRID MODE)

Original Application No. 985/2019

In Re : Water Pollution by Tanneries at Jajmau, Kanpur, Uttar Pradesh

WITH

Original Application No. 986/2019

In Re : Water Pollution at Rania, Kanpur Dehat & Rakhi Mandi, Kanpur Nagar, Uttar Pradesh

Date of hearing: 16.05.2023

**CORAM: HON'BLE MR. JUSTICE ADARSH KUMAR GOEL, CHAIRPERSON
HON'BLE MR. JUSTICE SUDHIR AGARWAL, JUDICIAL MEMBER
HON'BLE DR. A. SENTHIL VEL, EXPERT MEMBER**

Amicus Curiae: Ms. Katyayni, Advocate

Respondent: Mr. Pradeep Misra & Mr. Daleep Dhyani, Advocates for UPPCB

ORDER

The Issue

1. This order is in continuation of earlier orders of this Tribunal on two issues. First is grievance against failure to **scientifically handle and dispose of Chromium dumps at Rania, Kanpur Dehat and Rakhi Mandi, Kanpur Nagar which have been in existence since 1976** and have *inter-alia* resulted in contamination of ground water adversely affecting the health and depriving the inhabitants of access to drinking water (i.e. OA 986/2019). Second issue relates to **continuing water pollution by tanneries discharging untreated industrial effluents, containing toxic Chromium into the irrigation canal through inadequately functioning CETP at Jajmau, UP (i.e. OA 985/2019).**

Procedural History

2. The matter has been considered by the Tribunal in the last about four years inter alia vide orders dated 15.11.2019, 16.7.2020, 8.2.2021, 23.11.2021 and 23.09.2022. On first issue, it was found that though chromium dump site is in existence since 1976, no meaningful remedial action has been taken by the State to tackle such hazardous waste. This has affected the environment adversely, particularly the ground water. On the second issue, it was found that chromium contaminated effluent from tanneries is being discharged into river Ganga without proper treatment. On both counts the problem is still persisting in absence of adequate action on the part of the State authorities which calls for remedial action in mission mode, including fixing of responsibility of the erring officers for such continuing failure. We may refer to the orders passed earlier, dealing with the issue.

Order dated 15.11.2019

3. The status of compliance was reviewed on 15.11.2019 with reference to the report of the CPCB dated 30.10.2019 and earlier proceedings before this Tribunal. It was observed:

“11. It is undisputed that Chromium dumps containing toxic hexavalent Chromium (as mentioned in the report of the CPCB quoted above) has been in existence since 1976 and requisite steps have not been taken so far to dispose of the same as per mandate of law. Chromium is considered to be an environmentally hazardous element and classified as class-A human carcinogen.¹ Hexavalent Chromium Cr (VI) is toxic and the World Health Organization (WHO) has classified it as carcinogenic and can cause stomach ulcers and cancers and severe damage to kidneys and liver.² The industries responsible for generating the said dumps were closed in the year 2005. The SPCB has assessed liability of environmental compensation of

¹ <http://www.isca.in/rjcs/Archives/v7/i7/7.%20ISCA-RJCS-2017-024.pdf>

² https://www.who.int/water_sanitation_health/waterquality/guidelines/chemicals/chromium.pdf

Rs. 280.01 crore only on 24.10.2019. There is no explanation why no such step was taken against the said industries earlier. We may note that this Tribunal has been issuing directions for shifting of the Chromium dumps but the State of UP has failed to do so. The direction of this Tribunal has already been quoted above from the order dated 22.08.2019 (para 24). Such directions were also issued earlier vide order dated 13.07.2017.

12. **From the above, it is clear that there is failure on the part of State of UP and its authorities in disposal of the Chromium dumps which is hazardous to the public health and the environment and the proposal now mentioned in the report of the Chief Secretary, UP is for in-situ remediation though earlier stand of the State of UP was to shift the Chromium waste to the Treatment Storage and Disposal Facility (TSDF) for hazardous waste as per Hazardous Waste Management Rules, 2016.** The fact remains that the problem has not been tackled for the last 43 years and it has resulted in contamination of ground water affecting the health and life of the inhabitants and fauna. Compensation has been assessed only in the year 2019 without it being clear whether there is a chance of actual recovery of the same. There is no explanation for earlier inaction by the State of UP and the UPPCB.
13. For this failure, under the Public Trust Doctrine, the State is liable to deposit the said assessed amount in an ESCROW account for restoration of environment and the public health in the area. Such deposit may be made within one-month from today. The amount may be spent after preparation of an action plan by the District Magistrates and the SPCB with the approval of the CPCB. The ESCROW account will be operated by the concerned District Magistrate in terms of action plan. The State will be at liberty to recover the amount from the erring industries and/or from the erring officers who failed to take necessary action.

For measures to be adopted to utilize the amount, it will require a credible study of the health issues in the area. This may be done by an Expert Committee comprising representatives from (1) S.N. Medical College, Kanpur, (2) PGI Lucknow, (3) RML Lucknow and (4) a nominee of Secretary, Health, Ministry of Health, Govt. of India. The nodal agency will be the Principal Secretary, Medical & Health, UP.

14. With regard to supply of potable water in the affected areas, such supply should take care of not only drinking purposes but also other purposes. **It is well known that adverse effect on health is not only by drinking contaminated water but also on account of bathing or cooking and also on account of it being part of the food chain. It is necessary to put the concerned inhabitants in the area to notice of adverse consequences of use of contaminated water and placing the data of contents of water quality on website of the State. The affected area should also be delineated and put in public domain.**

15. *PWS must be established as is said to have already been directed by the State expeditiously positively from 01.03.2020. Since in Rakhi Mandi pipe carrying potable water is already available, such supply may be made operational positively by 15.01.2020 i.e. within two months, which is the timeline proposed by the State itself in its affidavit.*
16. ***With regard to illegal permission granted by the Principal Secretary, Urban Development on 08.08.2019 for release of large quantity untreated sewage directly into river Ganga on the ground of cleaning trunk sewer, the explanation furnished cannot be accepted as no assessment of pollution load and its constituents was made. It was not considered that the sewage/effluents had highly toxic Chromium content. Its impact on recipient water of river Ganga and the downstream inhabitants who will use such water was not considered. The action taken cannot by any standards be considered to be a responsible action of a welfare State and shows total apathy for the environment and the health of the inhabitants and the rule of law.***
Moreover, it is only after the order of this Tribunal that a decision has been taken to close operation of 122 tanneries which were discharging untreated industrial effluents with hazardous contaminants in irrigation channel through CETP and thereafter directly in the River Ganga as CETP did not have the requisite capacity. This action has been taken only on 01.10.2019. Tannery industries in India are contributing high Chromium contamination to the environment. These industries of India alone are reported to contribute about 2000-3000 tonnes of Chromium contamination to the environment in which Chromium concentration ranges from 2000- 5000 mg/L in the aqueous effluent.³
17. *The stand of the State of UP shows that it is being understood in certain quarters that during monsoon any pollution load, including sewage or any other polluting effluents can be discharged in the water bodies/rivers which is clearly against the mandate of Section 25 of the Water (Prevention and Control of Pollution) Act, 1974. **The CPCB may need to issue an appropriate direction to ensure that such illegality does not take place anywhere in the country.***
18. *The State of UP has to be held liable to pay compensation to the extent of Rs. 10 Crores for violation of law affecting the environment and public health for illegally permitting discharge of sewage and other effluent containing toxic Chromium directly into river Ganga. The quantum of compensation is being fixed having regard to the magnitude and nature of pollutant. **The report of the Chief Secretary in para 12 clearly accepts that the effluents of 122 operational tanneries now closed from 01.10.2019 was part of the discharge on account of stoppage of flow of effluents in CETP. Further***

³ Dhal B., Thatoi H.N., Das N. and Pandey B.D. (2013). Chemical and microbial remediation of hexavalent chromium from contaminated soil and mining/metallurgical solid waste: A review. *Journal of Hazardous Materials*, 250, 272-291

in para 13 it is stated that UP Jal Nigam was allowed to discharge effluents into river Ganga pending cleaning of trunk sewer and non-functioning of STP. Annexure 9 to the affidavit which is a report of the Principal Secretary, Urban Development mentions that the trunk sewer of dia. of 2100x2300mm was required to be cleaned which had capacity to carry 100 MLD sewage to cluster of STP of 205 MLD capacity. Main sewer line was damaged by tanneries mixing industrial waste into domestic waste which increased load for treatment on STP. This led to mixing of Chromium in sewage water rendering sludge unusable and harmful for the agricultural fields. Liability of any authority undertaking hazardous activity having potential for injury to environment and public health is well known.⁴ Principles for determining quantum are well settled.⁵ Compensation has to be approximate to the cost of restoration and where exact data is not available, broad approximation having regard to attending circumstances is permissible. We have fixed the quantum in the light of these well-known principles.

Even if adequate dilution was available, the pollution load that too loaded with toxic Chromium is undoubtedly bound to affect the water quality at one or other place and has potential to endanger the health and lives of people. The Principal Secretary, Urban Development had no legal jurisdiction to permit such illegality in violation of Section 25 of the Water (Prevention and Control of Pollution) Act, 1974.

The UPPCB, unfortunately, has not taken any action against such illegality and against polluting 122 tanneries for a long time for which the UPPCB has to be held liable to pay compensation of Rs. 1 Crore.

The UP Jal Nigam is also liable for such neglect as it released untreated large quantity of sewage containing toxic Chromium in river Ganga. UP Jal Nigam is held liable for environment compensation of Rs. 1 Crore. The said amounts may be deposited with the CPCB within one month from today which may be overseen by the Chief Secretary, UP. The State of UP will be at liberty to recover the amount from the erring officers, apart from taking appropriate disciplinary or other departmental action in accordance with law.

19. We may now sum up our directions as follows:

- i. The State of UP is held liable for failing to take any action for shifting of Chromium dumps at Rania and Rakhi Mandi which resulted in damage to the

⁴ M.C. Mehta vs. Union of India 1987 (1) SCC 395

⁵ Sterlite Industries (India) Ltd. v. Union of India (2013) 4 SCC 575 : ¶ 47, T.N. Godavarman Thirumulpad v. UOI & Ors. (2006) 1 SCC 1 : ¶ 1, Indian Council for Enviro-Legal Action & Ors. v. Union of India & Ors. (1996) 3 SCC 212 : ¶ 67, Vellore Citizens Welfare Forum v. UOI , (1996) 5 SCC 647 : ¶ 11 to 13, M.C. Mehta v. Kamal Nath (1997) 1 SCC 388 : ¶ 10 , Public Trust Doctrine, ¶ 24, M.C. Mehta v. UOI & Ors., W.P (C) No. 13029/1985 order dated 24.10.2017, MCD v. Uphaar Tragedy Victims Association (2011) 14 SCC 481 : ¶ 99, 100, Vadodra Municipal Corporation v. Purshottam v. Murjani & Ors. (2014) 16 SCC 14 : ¶ 17 and M. C. Mehta & Anr. v. Union of India (1987) 1 SCC 395 : ¶ 32.

environment and the public health for the period from 1976 till date. The amount of compensation in this regard is held to be the amount assessed by the UPPCB to be recovered from the erring industries. Till such recovery, the State itself must pay the amount by way of transfer to an ESCROW account. The amount is to be utilized for restoration of the environment and the public health in the area in the manner mentioned earlier. The State of UP is at liberty to recover the amount from the erring industries or erring officers as already mentioned in para 13 above.

- ii. The State of UP must take further steps for disposal of the hazardous Chromium dumps as per directions of this Tribunal dated 22.08.2019 quoted above, failing which it will be liable to pay compensation as mentioned in the said order.*
 - iii. State of UP is held liable to pay environmental compensation of Rs. 10 crores for damage to the environment for permitting discharge of untreated sewage containing toxic Chromium into river Ganga directly vide its order dated 08.08.2019. The State of UP is at liberty to recover the amount from the erring officers apart from taking action against the persons responsible in the manner as already mentioned in para 18 above. The UPPCB is held liable to pay sum of Rs. 1 crore for ignoring illegal discharge of sewage and other effluent containing toxic Chromium directly into river Ganga and taking action after a long time inspite of earlier proceedings before this Tribunal. UP Jal Nigam is held liable to pay sum of Rs. 1 crore for releasing untreated large quantity sewage containing toxic Chromium in river Ganga. These amounts may be deposited with the CPCB within one month which may be overseen by the Chief Secretary, UP. UPPCB is at liberty to recover the amount from the erring industries.*
 - iv. The State of UP may take steps for supply of potable water to the inhabitants of the area and other steps as already mentioned in paras 13 to 15 above.*
 - v. The Expert Committee in terms of para 13 above may conduct the health survey within three months.*
 - vi. CPCB may issue appropriate directions to ensure that no authority allows discharge of polluted sewage or polluted effluents directly into a water channel or stream in violation of law even in monsoon and also the standards for faecal coliform are duly maintained.*
- 20. Compliance report of the above directions may be filed by the Chief Secretary, UP before the next date by e-mail at judicial-ngt@gov.in.”*

Order dated 16.7.2020

4. The matter was then considered on 16.07.2020 in the light of the reports of the Chief Secretary, UP dated 04.02.2020 and 11.06.2020 and report of CPCB dated 14.07.2020 as follows:-

“1to3.....xxx.....xxx.....xxx

4. *The report of the Chief Secretary, U.P is that an action plan has been prepared for restoration of the environment and certain steps have been taken for supply of water to the inhabitants. Further report dated 11.06.2020 is that the matter of remediation is at the tender stage. The report of the CPCB is of a general nature.*

5. ***The fact remains that the chromium dump containing toxic chemicals has not been shifted to the TSDF as required under the law for which failure of the State of U.P is continuing inspite of repeated directions showing lack of sensitiveness on the part of the concerned officers. Hazard to public health and environment continues. The process of remediation can only start only after shifting of the waste to operational TSDF.***

6. ***Having regard to the seriousness of the consequences for continued delay on one pretext or other, we direct the Chief Secretary, U.P to ensure prompt action on priority basis in a time bound manner which may be personally monitored by the Chief Secretary, U.P. The Committee constituted by this Tribunal headed by Justice S.V.S. Rathore, former Judge of the Allahabad High Court may also oversee the compliance of this direction and give its independent report by e-mail at judicial-ngt@gov.in preferably in the form of searchable PDF/ OCR Support PDF and not in the form of Image PDF. It is made clear that the Tribunal may have no other option except to take coercive measure for any further default by the State of U.P. Compliance status as on 31.10.2020 may be reported to this Tribunal before the next date by e-mail at judicial-ngt@gov.in preferably in the form of searchable PDF/ OCR Support PDF and not in the form of Image PDF.”***

Order dated 8.2.2021

5. The matter was considered on 08.02.2021 in the light of report of the Oversight Committee dated 23.12.2020 and response of the State of U.P. dated 29.01.2021. The Tribunal found that in view of presence of mercury in the ground water there was need for cumulative impact study about the adverse effect on food safety and health. There was also need for multispecialty health facility for the affected inhabitants. Accordingly, the

Tribunal directed remediation of hazardous waste sites, including the additional sites pointed out in the report of the Committee, which was required to be monitored at the level of Chief Secretary in coordination with the Oversight Committee. It was further directed that water supply may be ensured to all affected areas and impact of contamination on food chain may be controlled. In case the compensation was not recovered from the violators, the liability was held to be of the State. The directions are reproduced below:

“10. It has also been pointed out that there are reports of mercury being present in ground water which necessitates cumulative impact study of the area in terms of possible adverse effect on food safety and health. There is also need for multispecialty health facilities for the affected inhabitants. Even though as per earlier health studies there is no adverse effect seen, in reality people are suffering from Cancer, Diabetes and other diseases on account of pollution. The study, thus, needs to be more credible.

*11. Accordingly, in the light of above discussion, **the State of UP may take further remedial action. Remediation of hazardous waste sites departmentally may be explored, as tender process has not succeeded in the last several years and the dump sites have been in existence for the last 45 years which continue to be hazard to the lives of the inhabitants. The additional sites pointed out above may also be remediated. This needs to be tackled on emergency basis at the highest level in the State and monitored at the level of the Chief Secretary and also the Oversight Committee. CPCB may be consulted in the process for the remedial action which should be as per rules. The action plan may also ensure provision for water supply to all affected areas. Further, it must be ensured that contamination does not affect the food chain. The stand of the State of UP with regard to compensation is not as per earlier order dated 15.11.2019, para 13. If the amount is not recovered from the companies for want of effective legal steps for recovery within reasonable time, liability of the State for its negligence and inaction will continue.***

*12. The report discussed above, only relates to scientific disposal of chromium dumps at three locations. **Nothing is mentioned about the second issue of continued pollution by tanneries as well as discharge of untreated sewage into the irrigation canals and drains at and around Jajmau. Requisite remedial action may now be taken expeditiously and a report furnished alongwith the further action taken report on remediation of dump sites with compliance status as on 30.06.2021, by 15.07.2021** by e-mail at judicial-ngt@gov.in preferably in the form of searchable PDF/ OCR Support PDF and not in the form of Image PDF.”*

Order dated 23.11.2021

6. The matter was then considered on 23.11.2021 in the light of status reports filed by the CPCB on 30.07.2021 with respect to inspection of Common Effluent Treatment Plant (CETP), Jajmau and Common Chrome Recovery Plant (CCRP), Jajmau on 30.06.2021 and 17.06.2021. Considering the said report, it was found that remedial action was required for scientific handling of the chromium dump. Further, steps were required for treatment of tanneries effluent before its discharge into the sewers and drains. The Tribunal accordingly constituted a five-member Committee under the Chairmanship of Chief Secretary, UP for remedial measures and directed filing of an action taken report before the Tribunal. Relevant extracts from the said order are reproduced below:

“xxxxxx.....xxx

8. *The response of the State PCB filed on 30.07.2021 is that UPSIDA has invited tender and appointed IIT Kanpur as consultant. It was found that actual volume of chromium at the site was much higher than mentioned earlier. Correct quantity is to be ascertained and then budgetary approval taken from the Government. Compensation has been assessed against six defaulting industries who have challenged their liability and the matter is pending consideration. Government has deposited a sum of Rs. 23.44 crores but the remaining amount has yet to be deposited. Piped water supply has been provided. Preliminary assessment has been made by the Expert Committee on the subject of health survey. Further survey is being conducted. Report of ground water analysis at 13 locations has been submitted in April 2021. The tanneries are now treating the effluents through the CETP. Action has been taken against the non-compliant units and compensation has been assessed, including the CETP operator. Functioning of STPs in Jajmau and Kanpur is being monitored.*

9. *Separate report of the Oversight Committee dated 12.11.2021 in continuation of earlier report dated 26.07.2021 mentions the progress.*

10. *From the report of the CPCB and the Oversight Committee dated 12.11.2021, it is seen that issue of remediation of chromium dump site is still hanging fire. It is now reported that actual volume of chromium waste is much higher than reported earlier. The affected area now found is 1,22,800 m³ as against the area earlier reported*

37,712 m³. Further, CETP and CCRP at Jajmau are non-compliant, calling for remedial action.

11. As already observed, there does not seem to be adequate seriousness on the part of the authorities in handling the grim situation which has been in existence since 1976 and is adversely affecting the health of the citizens. Further, discharge of toxic chromium is taking place in Ganga due to non-compliant CETP and CCRP for which prompt action is required, to be monitored at the level of Chief Secretary. First imperative is to collect complete information on number of wells contaminated as well as the locations where chromium dumps are existing in the form of a complete inventory. Next step is firm action plan. At the same time remediation work pending for such a long time is prompt need to protect health of citizens. Not realizing the seriousness, the issue appears to be stuck in identifying the executing agency. Having regard to seriousness of the matter, MoEF & CC needs to be involved for guidance and handholding.

12. On the issue of not having complete treatment of tanneries effluent, it is seen that the CCRP is being operated with limited Members with discharge by tanneries into existing sewers and drains. This shows poor consent monitoring mechanism on the part of State PCB. Tannery effluent and sewage effluent are reported to be mixed and utilised for irrigation (about 214 mld) having deleterious effect on soil, ground water and ultimately on crops grown. During the period when effluent is not utilized, the same may be going to Ganga. This situation needs to be checked and remedied. As noted in the order dated 15.11.2021, the State PCB has assessed compensation on account of continuing damage by the chromium dump at Rs. 280.01 crores as on 24.10.2019. Further, for discharge of untreated sewage of toxic chromium into Ganga, the State of UP has been held liable to pay compensation of Rs. 10 crores. UP Jal Nigam was also liable to pay compensation of Rs. 01 crore for discharge of untreated sewage. Compensation was liable to be paid by the State by way of transfer to an ESCROW account for restoration of the environment with liberty to recover the same from erring industries. As per report of the Oversight Committee, lifting of dump has not yet started while work of preventing discharge of effluent has commenced. State Government has so far deposited a sum of Rs. 23.44 crores. The State is liable to deposit the remaining amount and pendency of proceedings against erring industries cannot be a ground not to comply with the said order.

13. Accordingly, we find it necessary to constitute a five-member expert Committee under the Chairmanship of Chief Secretary, U.P. with nominees of MoEF&CC, CPCB, NMCG, UP PCB and District Magistrates of Kanpur Dehat and Kanpur Nagar. CPCB and the State PCB will be nodal agencies jointly. The Committee may meet within two weeks and prepare road map for remedial action. The Committee is free to coopt any other expert or institution. Cost of remediation will be first born by the State of UP with liberty to the State to recover the same from any other source of funding or from such erring industries/individuals as may be found responsible, as per law. The plan should result in remediation of dump sites as well as preventing further discharge of chromium or other effluents at Jajmau or other

such places in Ganga. An action taken report of compliance status as on 31.03.2022 be filed before the next date.”

Last Order dated 23.09.2022

7. By the last order, the Tribunal considered reports of the State PCB dated 30.07.2022, 18.08.2022 and 21.09.2022 and reports of Oversight Committee dated 22.04.2022 and 26.07.2022 and observed:-

“8. Report of State PCB dated 30.07.2022 is that the progress was reviewed at the level of the Chief Secretary in the matter of disposal of the chromium waste dumped. It was noted that e-tenders were invited, volume of the waste was ascertained, budgetary provision was made, environmental compensation was levied on six defaulting industries and thereafter notice was issued to five additional units, piped water supply was ensured for the inhabitants, steps were taken for maintenance of CETP, including levy of compensation on the CETP operator and for monitoring of status of tapping of drains and functioning of STPs. Action was also taken against the violating tannery units. Another CETP is being constructed.

9. In second report filed on 18.08.2022, the State PCB has submitted that rate for disposal of the waste was determined and disposal was to be through M/s U.P. Waste Management Project, Kumbhi, Kanpur Dehat. The waste was being covered by HDPE sheet by UPSIDA, so that no more contamination could be spread in rainy season.

*10. The third report filed on 21.09.2022 is that the Committee has finalized the rate as Rs. 11.583 per Kilogram for scientific and safe disposal of Chromium waste dumped at Khanchandpur, Rania, Kanpur Dehat through existing nearby TSDFs. The total cost for the disposal of approx. 85008 MT Chromium waste dumped at Khanchandpur Rania, Kanpur Dehat is estimated as approx. Rs. 98,46,47,664/-. The State Government has already released the fund of Rs. 48.44 Crore in the dedicated escrow account operated by District Magistrate, Kanpur Dehat for the same. **At present signing of MoU and placing the work order to both TSDF operators is in progress and thereafter lifting, treatment and scientific disposal of chromium waste is slated to begin from 15.10.2022, after the rainy season, so as to avoid leaching of chromium which may take place in the ground water. It will take atleast further six months after start of work for complete disposal of dumped hazardous waste to TSDF. At present the entire area is covered by polythene sheet.***

11. On the second issue of remedial steps against continuing pollution by discharge of effluents in drains and sewers by the CETPs at Jajmau, we may recall the observations in this Tribunal's order dated 22.07.2022 in OA No. 200/2014, M.C. Mehta vs. Union of India & Ors. Relevant extract from the said order is as follows:

“xxx.....xxx.....xxx

9. ... Non-compliance of CETP of the biggest tannery cluster at Jajmau is a serious issue for many years. Directions have been issued by CPCB to UPPCB dated 08.12.2021 for taking action which may include closure of member industries and levying environmental compensation. Similarly, CPCB has issued directions on 14.02.2022 and 10.01.2022 in respect of non-compliant CETP at Unnao and Banthar. CPCB needs to closely monitor with SPCBs about remedial action against non-compliant CETPs and member units, including action against those non-compliant with PETP standards. Since effluent of CETP at places like Jajmau goes to river Ganga during rainy days when not required for irrigation, industries need not to operate or reuse the diluted effluents back in process after nano-filtration etc., as required. ...”

12. We also note directions issued by CPCB on August 18/23, 2022 under Section 18(1) (b) of the Water (Prevention and Control of Pollution) Act, 1974 as follows:

“xxxxxx.....xxx

AND NOW, THEREFORE, in view of above observations and in exercise of the power conferred under section 18 (1) (b) of the Water (Prevention & Control of Pollution) Act 1974, you are here by directed to take appropriate measures for compliance of following directions in a time bound manner:

1. The compliance of the norms is mandatory and tanneries & CETPs shall not be allowed to operate with repeated non-compliance till completion of CETPs upgradation work take place.
2. Uttar Pradesh Pollution Control Board (UPPCB) shall issue directions following applicable procedure under section 33A of the Water (Prevention and Control of Pollution) Act, 1974 to all the Tannery Units operating in three tannery clusters of Kanpur-Unnao region in Uttar Pradesh (CETP members and standalone, both) to take action for compliance of the following pollution control measures:
 - a. Cleaner technology options such as mechanical desalting, chilled/brine hides, enzyme-based low sulfide un-hairing, recycling of liming liquor, pickle-less tanning shall be implemented in three tannery clusters in organized manner through proposed charter and shall be made a part of consent mechanism.
 - b. Tannery Units shall prepare and secure the execution of their individual action plan for implementation of applicable/feasible cleaner technology/waste minimization options in their manufacturing processes including upgradation of PETP/ETP in a time bound manner and their

individual action plans shall be submitted to UPPCB for approval within 30 days. The "Charter for Water Recycling and Pollution Prevention in Tanneries" placed at Annexure-I may be referred for preparation of action plans. Action plans should also include monthly/quarterly monitorable targets.

- 3. UPPCB shall facilitate for setting up of common facility for chilled/brine preserved hides by slaughter houses and major suppliers.*
- 4. UPPCB shall validate and approve the action plan submitted by individual tannery units along with the plan for setting up of common facility for chilled/brine preserved hides and submit a comprehensive area specific plan for all three tannery clusters to CPCB within 45 days.*
- 5. UPPCB shall monitor the progress of the implementation of the action plans. In case tanneries/ CETPs fail to comply with their monitorable targets of action plans and CETPs/PETPs/ETPs are observed non-compliant during inspections, appropriate action may be taken against them including levying of environmental compensation which may be linked with the progress of the action plans of the individual tanneries/ CETPs. UPPCB shall also withdraw the 'Consent to Operate' issued to the tannery units in case of significant violation/ non-compliance/ non-adherence with the time schedule as prescribed under the action plans. The quarterly action taken report/progress report shall be forwarded to CPCB."*

13. Thus, on both the issues, some progress has been achieved but still chromium dump remains and effluent continues to be let out through drains and CETP. While rate for lifting of waste from Khanchandpur and waste stabilization/immobilization have been finalized, plan is required for remediation of contaminated groundwater. This may involve estimating cost, identifying executing agency and time plan. It is reported that there is no waste at Rakhi Mandi. It appears that chromium has been spread in large area with high concentration. As per report of UPPCB dated 18.08.2022, a study conducted jointly by CPCB and UPPCB reveals as under:

*"11. That UPPCB and CPCB have conducted the joint sampling on 16.06.2022 from the dump site to assess the Hexavalent chromium concentration present in waste dumped at Khanchandpur Rania, Kanpur Dehat. 44 nos. of waste samples were collected. **Out of 44 samples, 33 grab and 11 composite samples were collected. As per analysis reports. The observed average Cr (VI) in TCLP is 155mg/L (approx.) ranging from 99 mg/L to 214 mg/L. Total area of the dumped waste impacted with hexavalent chromium has been estimated as 43,180 m². Copy of rapid assessment report of CPCB is enclosed herewith and marked as Annexure no.- 6.**"*

14. Thus, it is clear that health risk remains on account of delayed inaction. UPPCB with other concerned agencies have to ensure that no waste is lying at any other site.

15. We may consider the situation of pollution of River Ganga due to untreated/partially treated effluents (sewage and industrial) being discharged thorough irrigation canal at Jajmau. The status reported by UPPCB on 30.7.2022 with reference to steps taken on this aspect are reproduced below:

“24. That it is further submitted that the CETP at Jajmau, Kanpur is having capacity of 36 MLD to treat (combined effluent in ratio of 1:3) 9 MLD Tannery effluent & 27 MLD domestic sewage effluent. All the tannery units discharge primarily treated effluent through conveyance channel, which is further pumped through 4 pumping stations into 36 MLD CETP designed to treat mixed effluent consisting of 9 MLD tannery effluent and 27 MLD domestic sewage.

That the maintenance of conveyance channel, pumping stations and operation of CETP is done by Ganga Pollution Control Unit, U.P. Jal Nigam, Kanpur. The treated CETP effluent gets further diluted with treated effluent of 130 MLD, 43 MLD and 5 MLD STP at the premises of CETP Jajmau before finally discharged into irrigation channel for irrigation. The STPs are also operated and maintained by operator M/s Kanpur River Management Pvt Ltd, Kanpur selected by UP Jal Nigam.

That it is further submitted that there is no separate sewerage network in Jajmau area, where all the tannery units are situated, so untreated domestic sewage generated from Jajmau area also gets mixed with industrial effluent goes through conveyance channel into CETP.

That, it is submitted that keeping the above aspect into consideration, U.P. Pollution Control Board has taken following measures to reduce the chances of overflow from pumping stations as well as excess effluent going into CETP.

- a. U.P. Pollution Control Board issued guidelines vide letter dated 08.06.2019 to assess the infrastructure of tannery units according to their permitted capacity in order to restrict tannery units to operate as per their permitted capacity.
- b. The process of tanning operation in tannery unit is such that there is no continuous flow generation from the process and it is more in the nature of batch process, so tannery effluent generated from tannery units gets discharged after accumulation of effluent for 2 to 3 days. This in turn results into excess flow generated from the tannery units at a particular point of time than the permissible discharge calculated on Kl/day basis, whenever it is discharged after completion of batch process. U.P. Pollution Control Board restricted the

overall capacity of each tannery unit first upto 80% of its consented capacity in order to reduce the chances of excess flow going to pumping stations and subsequently to CETP inlet. But this step was also not found adequate as excess flow was still noticed at each pumping stations and CETP inlet due to inseparable domestic effluent up to 10.02 MLD, as well as batch process of tannery units. Therefore, U.P. Pollution Control Board restricted the overall processing capacity of each tannery unit further to 50% of its consented capacity in order to reduce the chances of excess flow going to pumping stations and subsequently to CETP inlet.

- c. U.P. Pollution Control Board implemented roster of operation of tannery units attached to different pumping stations. As per this roaster, tannery units attached to pumping station 1 and 3 are operated for first 15 days of the month and tannery units attached to pumping station 2 and 4 are operated for next 15/16 days of the month.*
- d. Each tannery unit has installed flow meter at the outlet of PETP with connectivity to CPCB and UPPCB server.*
- e. U.P. Pollution Control Board is regularly monitoring the effluent quality of different conveyance channels before reaching pumping stations and it has been directed that if any default is noticed then coercive action will be taken against all the tannery units existing in upstream of that location.*
- f. Surprise checks are conducted by UPPCB and Inter Departmental Teams as well as Third Party Institutions like IIT, HBTU, PCCRI, NEERI also conduct the surprise inspections from time to time and actions are taken against the defaulter units.”*

16. *On the delayed project for setting up of CETP at Jajmau, the report states as:*

“25. That it is also relevant to submit here that in order to improve the treatment capacity of existing tannery units of Jajmau, Kanpur, Central Government has taken effective initiation and for Jajmau Tannery cluster, National Mission for Clean Ganga, New Delhi has approved 20 MLD CETP (dilution based) worth Rs. 554 Crore, as per Hon'ble NGT direction. The proposal was consented by the tannery association and the Government of Uttar Pradesh. The member tannery units have committed to contribute Rs. 25 crore and has proposed to bear the estimated cost of Rs. 137 Crore for upgradation of their Primary Effluent Treatment Plants (PETPs). Rest 392 Crore will be borne by National Mission for Clean Ganga (NMCG). The Special Purpose Vehicle "Jajmau Tannery Effluent Treatment

Association (JTETA)" of member tannery units and district administration headed by District Magistrate, Kanpur Nagar has been formed.

26. That the Jajmau Tannery Effluent Treatment Association (JTETA) has allotted the work of construction of CETP to M/s VA Tech Wa Bag Ltd. after finalizing the tender with approval from NMCG. NMCG has also appointed construction engineer to supervise the on going work and estimated time of commissioning is 13-08-2023. The new CETP is being constructed in the same premises of existing CETP and the treated effluent is proposed to be used for irrigation after dilution with treated STP effluent. After the construction of new 20 MLD CETP & allied structures like closed conveyance system to carry primarily treated effluent from member tannery units to CETP & CCRP and upgradation of existing PETP by tanneries, complete elimination of accidental discharge in river Ganga would be ensured.

27. That the member tannery units have also submitted undertaking for upgradation of PETP before commissioning of the new CETP.

28. That it is submitted that every member unit has also to contribute its share towards construction of new 20 MLD CETP. U.P. Pollution Control Board has issued closure order against 121 member tannery units who have not deposited their installments as per schedule, on the recommendation of (SPV) Jajmau Tannery Effluent Treatment Association (JTETA) headed by District Magistrate, Kanpur Nagar."

17. In view of above, it is patent that:

- i. **Contaminated sites need to be remediated in terms of removing of the waste and transporting to TSDF, followed by due stabilization/immobilization.**
- ii. **Contaminated chrome bearing groundwater needs to be remediated by appropriate methodology and plan executed in a time bound manner. UPPCB and CPCB may immediately find out suitability of setting up of pilot plant model to treat chromium by way of setting up of RO/Ultra/Nano – Filtration process and application of Biotechnology to remove chromium as being done at other places for removal of arsenic/fluoride. This may be done in next three months and the cost of such demonstration/model project may be borne through EC funds of the UPPCB.**
- iii. **CPCB needs to get its directions dated 18/23.08.2022 implemented on priority apart from action referred to in Para 11(1) above and also on other issues in a time bound manner.**
- iv. **UPPCB needs to monitor remedial measures to prevent pollution from tanneries in realistic manner issuing guidelines, roaster operations, sealing of drums/pedals, cutting down the**

production capacities, closing the tanneries till compliance is achieved.

- v. NMCG needs to ensure that CETP at Jajmau is completed as already long delay has taken place.***

18. In view of above, let further action be taken expeditiously and report about compliance status as on 31.01.2023 filed by the State of UP on or before 15.02.2023 by e-mail at judicial-ngt@gov.in preferably in the form of searchable PDF/OCR Support PDF and not in the form of Image PDF. CPCB and NMCG may also file their respective reports about compliance status in the like manner.”

Compliance status as presented for consideration after last order

8. In pursuance of above, reports have been filed by State PCB (21.02.2023) and Oversight Committee (22.02.2023, 24.03.2023 and 09.05.2023).

9. The report of the State PCB is that MoUs have been executed between the TSDF operators - M/s U.P. Waste Management Project & M/s Bharat Oil and Waste Management Ltd., Kumbhi, Kanpur Dehat and Additional District Magistrate (Admin) Kanpur Dehat on 05.11.2022 and work orders have been issued to the TSDF operators by Additional District Magistrate (Admin) Kanpur Dehat on 15.11.2022 for lifting, treatment and scientific disposal of chromium waste as per the rate decided by the committee constituted by Environment, Forest and Climate Change Department Government of U.P. Both the TSDF operators were directed to start the work from 17.11.2022. Both the TSDF operators started the work regarding completing the necessary formalities before start of the work such as construction of tyre washing platform, dust suppression arrangements, arrangement of weigh bridge, route chart for transportation of Chromium waste from dump site to TSDF facility, Demarcation of site for lifting the waste, fencing of the site by green net, installation of PTZ camera with connection to UPPCB's monitoring portal and construction of site office etc. It has been further stated:-

- “5. That it is submitted that a meeting of Committee and Sub Committee was again held on 22.11.2022 in hybrid mode to review the progress regarding completion of necessary formalities and both the TSDF operators were directed to complete all the formalities as per MoU at the earliest. Further a meeting with all the stakeholders was held under chairmanship of ADM (Admin), Kanpur Dehat on 06.12.2022 to further review the progress, in which officers of weight and Measure Department, Toll Plaza officers of Kanpur-Itawa Highway, Public Work Department, Regional Office, U.P. Pollution Control Board and both the operators of TSDFs. In the meeting both the operators have informed that they have completed all the formalities.
6. That it is submitted that lifting of Chromium waste from the site to TSDFs started on 15.12.2022 and it is going on. Committee has ensured the scientific disposal of waste and suitable SoP has been devised for the same. At the TSDFs, the weight of chromium waste is again measured and stored separately before stabilization. **After stabilization the sample of the stabilized waste is collected and sent to Laboratory of Central Pollution Control Board, Delhi for TCLP test and after the results are found as per the prescribed standards for disposal in SLF, the operators are allowed to dispose in the SLF. The quantity of Chromium waste lifted from the site is also monitored through a mobile app developed by U.P. Pollution Control Board. Both the TSDF operators have lifted approx. 28596 MT chromium waste from dump site till 20.02.2023.**
7. That Hon'ble NGT has directed UPPCB and CPCB to find out suitability of setting up of pilot plant model to treat chromium by way of setting up of RO/Ultra/Nano Filtration process and application of Biotechnology to remove chromium as being done at other places for removal of arsenic/fluoride. That it is submitted that in compliance of above direction a meeting was held through hybrid mode on 11.01.2023 regarding remediation of chromium contaminated ground water at Khanchandpur Rania, Kanpur Dehat. During meeting it was also informed that ground water remediation technology suggested in earlier approved DPR under NCEF project was pumped and Ex-situ treatment. **Further strategy for remediation can be decided after present condition of ground water quality and extent/level of contamination at Chromium contaminated sites at Rania, Kanpur Dehat after monitoring of ground water from old /existing monitoring wells.** Minutes of the first meeting held on 11.01.2023 and communicated vide letter dated 06.02.2023 are being attached herewith and marked as **Annexure No.-3**
8. That it is submitted that in the State of Uttar Pradesh, **similar nature of ground water contamination containing hexavalent chromium is being remediated at Lohia Nagar, Ghaziabad through pump and treat method along with combination of treatment through bio-culture/micro**

organisms. So, technology already exists. A brief status regarding the technology is attached herewith and marked as **Annexure No. -4**

9. That it is submitted that U.P. Pollution Control Board had imposed Environmental Compensation of Rs. 280.01 Crore against 08 defaulting industries on 14.04.2022. Hon'ble NGT has revised the amount of Environmental compensation imposed by UPPCB, considering the appeal filed by industries vide order dated 05.12.2022, 15.12.2022, & 16.01.2023 in Appeal Nos. 14, 15, 16, 17 and 18 of 2020. The detail of revised Environmental Compensation is as follows :

Sr.No.	Name & Address of industry	Environmental compensation amount imposed by the State Board (Rs.)	Revised environmental compensation amount by Hon'ble NGT (Rs.)
1	Chandani Chemicals Pvt. Ltd. Khanchandpur , Rania, Kanpur Dehat	27,40,62,295.00	23,09,31,000.00
2	HeilgerChem Pvt. Ltd. VIII.- Chiraura, Raipur, Rania, Kanpur Dehat	39,62,13,484.00	25,52,34,375.00
3	Rukmani Chemicals Pvt. Ltd. Rania, Kanpur Dehat	46,89,51,039.00	13,69,56,375.00
4	Amelia Textiles & Chemicals Pvt. Ltd. 23, Khanchandpur, Rania, Kanpur Dehat	14,61,66,558.00	11,92,50,000.00
5	Waris Chemicals Pvt Ltd. Khanchandpur, Rania, Kanpur Dehat	44,34,53,042.00	25,39,68,750.00
6	Cerulean Chemicals Pvt. Ltd. Khanchandpur, Rania, Kanpur Dehat	87,69,99,345.00	-
7	Kaleena Chemicals, Pvt Village Malo, GT Road, Chaubepur, Kanpur Nagar	18,02,72,088.00	-
8	Bharat Chemical Udhyog C/O Smt Tarannum Fatima W/o Shri Sahabuddin and Shri Sahabuddin S/o Mohd Nafees, 39/72, Majeed Ahmad Road, Kanpur Nagar and ShriIftekharul Amin, ShriIqbal Ahsan, ShriWaqarul Amin	1,40,07,150.00	-

10. That it is submitted that U.P. Pollution Control Board has also given direction vide letter dated 07.02.2023 to the concerned industries to deposit the environmental compensation as per the orders of Hon'ble NGT dated 05.12.2022, 15.12.2022, & 16.01.2023 in Appeal Nos. 14, 15, 16, 17 and 18 of 2020 with copy to District Magistrate, Kanpur Nagar for necessary action. The environmental compensation has not been deposited by any industry till now. The copies of letter dated

07.02.2023 are enclosed herewith and marked as **Annexure No.-5**

11. ***That it is submitted that U.P. Pollution Control Board is in the process to invite EoI also from various stakeholders to submit draft pilot plant project report for deciding possible technologies for ground water remediation at Khanchandpur Rania, Kanpur Dehat. That it is submitted that the work of ground water remediation could only be started after removal of lying chromium waste and stabilization of contaminated soil, so that their effect is ruled out from ground water. Latest ground water analysis is also required to be carried out to ascertain the extent of contamination of ground water in that area.***
12. *There are 400 tanneries in Jajmau Kanpur which are established in approximately 7.431 sq. kilometre area within 50 meter to 1.0 Km distance from river Ganga. The smallest tannery is of 100 sq. meter area and largest is of 34500 sq. meter area. The tanneries are established since 1946 at Jajmau. Most of the tanneries are small scale tanneries and have only installed primary effluent treatment plant and the treated effluent is being treated through a common effluent treatment plant installed by State Government through indodutch assistance in 199293. That it is further submitted that there is no separate sewerage network in Jajmau area, where all the tannery units are situated, so **untreated domestic sewage generated from Jajmau area also gets mixed with industrial effluent which goes through conveyance channel into CETP.***
13. ***That it is submitted that the CETP at Jajmau, Kanpur is having capacity of 36 MLD to treat (combined effluent in ratio of 1:3) 9 MLD Tannery effluent & 27 MLD domestic sewage effluent. All the tannery units discharge primary treated effluent through conveyance channel, which is further pumped through 4 pumping stations into 36 MLD CETP designed to treat mixed effluent consisting of 9 MLD tannery effluent and 27 MLD domestic sewer.***
14. *That the maintenance of conveyance channel, pumping stations and operation of CETP is done by Ganga Pollution Control Unit, U.P. Jal Nigam, Kanpur. The **treated CETP effluent gets further diluted with treated effluent of 130 MLD, 43 MLD and 5 MLD STP at the premises of CETP Jajmau before finally discharged into irrigation channel for irrigation. The STPs are also operated and maintained by operator M/s Kanpur River Management Pvt Ltd, Kanpur selected by UP Jal Nigam.***
15. ***That, it is submitted that in order to control industrial pollution from tannery sector in Jajmau area, State Government has devised a short term and a long term***

plan so that pollution of river Ganga due to effluent from tanneries is completely stopped.

- A.** That as a **Short Term Plan**, the strategy adopted is the improvement in compliance through enforcement measures as detailed below:
- a. That U.P. Pollution Control Board issued guidelines vide letter dated 08.06.2019 to assess the infrastructure of tannery units according to their permitted capacity in order to restrict tannery units to reduce the chances of overflow from pumping stations as well as excess effluent going into CETP.
 - b. That it is submitted that UPPCB has sealed 169 drums & 47 Paddles in 92 tannery units associated with Pumping Station-03 (Wajidpur), 54 drums & 12 Paddles in 19 tannery units associated with Pumping Station-04 (Budhiyaghat), 13 drums & 07 Paddles in 12 tannery units associated with Pumping Station-01 (Chhabilepurwa) to control excess flow generation.
 - c. That U.P. Pollution Control Board restricted the overall processing capacity of each tannery unit further to 50% of its consented capacity in order to reduce the chances of excess flow going to pumping stations and subsequently to CETP inlet.
 - d. That U.P. Pollution Control Board implemented roster of operation of tannery units attached to different pumping stations. As per this roster, tannery units attached to pumping station 1 and 3 are operated for first 15 days of the month and tannery units attached to pumping station 2 and 4 are operated for next 15/ 16 days of the month.
 - e. That U.P. Pollution Control Board has informed that Surprise checks are conducted by UPPCB and Inter Departmental Teams as well as Third Party Institutions like IIT, HBTU, PCCRI, NEERI from time to time and actions are taken against the defaulter units as well as operator of CETP. The details of action taken in form of closure of tannery units is enclosed herewith and marked as **Annexure No. - 6.**
 - f. That U.P. Pollution Control Board has imposed the environmental compensation against the defaulting units from time to time. Environmental Compensation of Rs 2.81 Crore imposed against 259 defaulter tannery units, out of which Environmental Compensation of Rs. 1.22 Crore has been deposited by 240 tannery units of Jajmau Kanpur. Recovery Certificate for recovery of Environmental Compensation has been issued by District Administration against 19 units.

- g. That U.P. Pollution Control Board has prosecuted 65 defaulter tannery units of Jajmau Kanpur under section 41/44 of Water (Prevention and Control of Pollution) Act 1974 in Special Court (Air and Water Pollution Control) Lucknow. Out of which 36 cases have been decided and conviction is done in 26 cases.
- h. That U.P. Pollution Control Board has imposed Environmental Compensation against the operator of CETP and STPs at Jajmau Kanpur i.e. Ganga Pollution Control Unit, U.P. Jal Nigam, Kanpur / M/s Kanpur River Management Pvt Ltd, Kanpur from time to time on the basis of different defaults. The detail of imposition of Environmental Compensation is enclosed herewith and marked as **Annexure No. -7**.

B. That as a **Long Term Plan**, the strategy adopted involves construction of a new CETP and upgradation of existing PETPs along with adoption of cleaner technology.

That it is submitted that in order to improve the treatment capacity of existing tannery units of Jajmau, Kanpur, Central Government has taken effective initiation and for Jajmau Tannery cluster, **National Mission for Clean Ganga, New Delhi has approved 20 MLD CETP (dilution based) worth Rs. 554 Crore, as per Hon'ble NGT direction. The proposal was consented by the tannery association and the Government of Uttar Pradesh. The member tannery units have committed to contribute Rs. 25 crore and has proposed to bear the estimated cost of Rs. 137 Crore for upgradation of their Primary Effluent Treatment Plants (PETPs). Rest 392 Crore will be borne by National Mission for Clean Ganga (NMCG). The Special Purpose Vehicle "Jajmau Tannery Effluent Treatment Association (JTETA)" of member tannery units and district administration headed by District Magistrate, Kanpur Nagar has been formed.**

That the Jajmau Tannery Effluent Treatment Association (JTETA) has allotted the work of construction of CETP to M/s VA Tech Wa Bag Ltd. after finalizing the tender with approval from NMCG. NMCG has also appointed construction engineer to supervise the ongoing work and estimated time of completion is 13.08.2023. The new CETP is being constructed in the same premises of existing CETP and the treated effluent is proposed to be used for irrigation after dilution with treated STP effluent. NMCG has also reviewed the progress of installation of 20 MLD CETP on 06.02.2023 with concerning stakeholders.

After the construction of new 20 MLD CETP & allied structures like closed conveyance system to carry primarily treated effluent from member tannery units to CETP & CCRP and upgradation of existing PETP by tanneries, complete elimination of accidental discharge in river Ganga would be ensured.

16. That it is submitted that Central Pollution Control Board has prepared a "Charter for Water Recycling and Pollution Prevention in Tanneries" in order to reduce the TDS in effluent generated from the tannery units by adopting cleaner technology and to minimize the quantity of waste. **Central Pollution Control Board has given various directions to U.P. Pollution Control Board vide letter dated 18/23.08.2022. In compliance of above, UPPCB has issued directions vide letter dated 02.09.2022 under section 33-A of Water Act 1974 to all tannery units, for adopting cleaner technology to minimize effluent quantity and use of chemicals as per the "Charter for Water Recycling and Pollution Prevention in Tanneries" in time bound manner and submit action plant for the same. Central Leather Research Institute (CLRI) has given license to 43 tannery units of Jajmau for use of Technology regarding waterless chrome tanning process. In continuation of adopting cleaner technology, M/s Solidaridad Asia (having technical collaboration with Neatherlands Enterprise Agency) has installed brush type desalting machine in 18 tannery units, fitted water meter in 30 drums of 20 tannery units, installed drum screen in 01 tannery unit, Screen Bar in 25 tannery units and tallow plant in 01 tannery unit of Jajmau, Kanpur. Besides above 2500 square feet tiles have been constructed from lime sludge obtained from 05 tannery units of Jajmau, Kanpur. Upgradation of PETP in 10 tannery units is also proposed by Solidaridad Kanpur Centre in Jajmau, Kanpur. The member tannery units have also submitted undertaking for upgradation of PETP before commissioning of the new 20 MLD CETP.**

That it is submitted that a workshop at Kanpur Nagar has also been organized by CPCB and UPPCB with stakeholders and cleaner technology service providers regarding adoption of cleaner technology on 07-08 February 2023."

10. Report of the Oversight Committee dated 22.02.2023 is as follows:-

Sl No	Directions issued by the Hon'ble NGT	Action taken
1.	Contaminated sites need to be remediated in terms of removing of the waste and transporting to TSDF, followed by due stabilization/ immobilization.	Scientific disposal of the chromium dump has started on 15.12.2022 in Khanchandpur Rania, Kanpur Dehat through two TSDFs, namely, (i) M/s Uttar Pradesh Waste Management Project, a Division of Re Sustainability Ltd, formally known as Ramky Enviro Engineers Ltd, & (ii) M/s Bharat Oil and Waste Management Ltd, Kumbhi, Akbarpur, Kanpur Dehat having processing capacities of 685 Mt/d and 100 Mt/d respectively. Quantities remediated till 31st January 2023 through these two TSDFs are (i) 12126.27 Mt and (ii) 1496.97 Mt respectively (total 13623.24 Mt).The entire quantity of waste is scheduled to be disposed of by May, 2023.

		<i>The work-plans submitted by the TSDFs are being annexed herewith as Annexure-1.</i>
2.	<i>Contaminated chrome bearing ground water needs to be remediated by appropriate methodology and plan executed in a time bound manner. UPPCB and CPCB may immediately find out suitability of setting up of pilot plant model to treat chromium by way of setting up of RO/ Ultra/ Nano-Filtration process and application of Biotechnology to remove chromium as being done at other places for removal of arsenic/fluoride. This may be done in next three months and the cost of such demonstration/ model project may be borne through EC funds of the UPPCB.</i>	<i>The UPPCB has informed that it is in the process of inviting EoI from various stakeholders for the submission of draft pilot plant project report with a view to deciding the possible technological remedies for ground water remediation at Khanchandpur, Rania, Kanpur Dehat. The work of ground water remediation could be started only after removal of lying chromium waste and stabilization of contaminated soil.</i>
3.	<i>CPCB needs to get its directions dated: 18/23.08.2022 implemented on priority apart from action referred to in Para 11(1) above and also on other issues in a time bound manner.</i>	<i>The UPPCB has informed that:</i> <i>(a) it has issued a letter dated 02.09.2022 with directions under Section 33-A of the Water Act 1974 to all tannery units for adopting cleaner technology with a view to minimize effluent quantity and use of chemicals as per the "Charter for Water Recycling and Pollution Prevention in Tanneries" in a time bound manner. Subsequently, directions for closure and disconnection of electricity of 213 tannery units were issued on 15.09.2022. However, no information on actual closure of these units and disconnection of electricity has been furnished.</i> <i>(b) Consequent upon issue of the letter dated 02.09.2022, Environmental Compensation has been imposed on the following defaulters on different dates:</i> <i>(i) Rs. 48.375 lacs on GPCU, UP Jal Nigam (Urban) for the CETP Jajmau, Kanpur Nagar;</i> <i>(ii) Rs. 21.0 lacs on the KRML Agency for the STP Jajmau, Kanpur Nagar; and</i> <i>(iii) Again, Rs. 14.20 lacs on the KRML Agency for the STP Jajmau.</i> <i>However, information on recovery of ECs imposed in the above and previous cases has not been shared.</i> <i>(c) The Central Leather Research Institute (CLRI) has granted license to 43 tannery units at Jajmau for use of Technology for waterless chrome tanning process. M/s Solidaridad Asia (in collaboration with Neatherlands Enterprise Agency) has installed brush type desalting machine in 18 tannery units, fitted water meter in 30 drums of 20 tannery units, has installed drum screen in 01 tannery unit, screen bar in 25 tannery units and tallow plant in 01</i>

		<i>tannery unit at Jajmau, Kanpur. In addition, 2500 square feet tiles have been constructed from lime sludge obtained from 05 tannery units of Jajmau, Kanpur. Upgradation of PETP in 10 tannery units is proposed by M/s Solidaridad, Kanpur Centre in Jajmau, Kanpur. The member tannery units have also submitted undertaking of upgradation of PETP before commissioning of the new 20 MLD CETP.</i>
4.	<i>UPPCB needs to monitor remedial measures to prevent pollution from tanneries in realistic manner issuing guidelines, roaster operations, sealing of drums/pedals, cutting down the production capacities, closing the tanneries till compliance is achieved.</i>	<i>The UPPCB has informed that in order to control industrial pollution from tannery sector, it has issued several instructions on roaster operations, sealing of drums/pedals, cutting down the production capacities and, closure orders against the non-compliant tanneries. It also undertakes enforcement measures regularly. However, as per the data available on the UPECP web portal regarding water quality of river Ganga at Kanpur, its upstream and downstream continues to remain in D Category. Further, the water quality of river Ganga at downstream of Kanpur is worse than the upstream indicating that these units and other wastes add to the pollution of the river Ganga in this stretch. For details the Annexure-2 may kindly be referred.</i>
5.	<i>NMCG needs to ensure that CETP at Jajmau is completed as already long delay has taken place.</i>	<i>As per information received from the NMCG 66.95% construction work of the 20 MLD CETP at Jajmau Kanpur, has been completed till 31st January 2023. The targeted time for completion has been reported as August 2023. Having regard to the pace of work in past about 18 months, it does not appear to be realistic. Till June, 2021, the progress was about 30%. In February, 2022 it became 43.5%, in June, 2022 it was 54% and in January, 2023, it is reported as 67%. On each of these reporting dates, except the last one, an unrealistically closer target date was given. This has been pointed out in earlier reports of this Committee. In last 07 months, only 13% progress has been made. At the present pace, it appears difficult to complete the project by August, 2023.</i>

11. Report of the Oversight Committee dated 09.05.2023 mention the compliance states as follows:-

“Status of Compliance:

Progress regarding various directions issued by the Hon’ble NGT is same as reported in the previous reports except on direction no. 01 where it has been reported that the M/s Uttar Pradesh Waste Management Project has lifted 72,283 MT and treated 57,744.645 MT out of 80,000 MT of allocation of the waste and M/s Bharat Oil and Waste Management Ltd has lifted 4962 MT and treated 2898.54 MT out of 5000 MT of allocation of waste. Considering the pace of work and the quantity of waste remediated till date, scientific remediation and disposal of the entire waste, appears to be achievable by the target date.”

12. We have further considered the matter. Before proceeding further, we may also note contents of letter dated 27.09.2022 received from Ms. Pratiba Shukla, Minister of State, Mahila Kalyan Bal Vikas and Pushtahar as follows:-

"I refer the letter No. H82151/C-2/NGT-46/2021 dated 21/09/2022 submitted by UP Pollution Control Board (UPPCB) to the Hon'ble NGT in case of hazardous dump existing at Rania, Kanpur Dehat which is incorrect and misleading as per the following points :-

- (i) As per the 3rd para of said letter of UPPCB "At present the entire area is covered by Polythene Sheet", the fact is that not more than 50% area is covered upto 26/09/2022 when the rainy season is going to over. In this connection, the entire money of the Govt. i.e. about to Rs.3 crore approx is to be waste that too when the shifting of dump to begin from 15/10/2022 as per the para (iii) of the said letter dated 21.09.2022 of UPPCB. In this connection, a video of the dump shooted on 26/09/2022 is being sent to you in pen drive for your information and perusal.*
- (ii) The UPPCB has exempted 3 Industries from the environmental compensation namely (1) Unichem India, Choubepur, Kanpur Nagar (2) Khanna Vivek Chemicals (P) Ltd. Panki, Kanpur Nagar and (3) IGS Chemicals (P) Ltd., Mandhna, Kanpur Nagar on the basis of hazardous waste generated 35% to 40% of the total chromite ore (basic raw material containing Cr₂O₃) consumed by the industries.*
- (iii) As per the above ratio accepted by the UPPCB in case of the above 3 industries should have also been followed in case of other basic chrome sulphate industries operating in Kanpur Nagar, Kanpur Dehat and Unnao Districts.*
- (iv) As per the information to me the 2 industries out of 8 have been storing their hazardous waste within the factory premises in the pucca pit and sheds since inception. They are Waris Chemicals Pvt. Ltd and Heilger Chemicals Pvt. Ltd. In my opinion, They would have also been given a chance to dispose off the hazardous waste to TSDF and should have been exempted from the environmental compensation as per the above 3 industries mention in point No.(ii) if the hazardous waste is found of the quantity already accepted in the case of other 3 exempted industries.*
- (v) As per the information to me, the total Chromite ore used by all the 8 industries in question is about 57881 MT (Cerulin Chemicals Pvt. Ltd.- 19,484.438 MT, Waris Chemicals-6005.00 MT, Chandni Chemicals Pvt. Ltd.-*

8798.00 MT, Amelia Textile & Chemicals Pvt. Ltd.-3833 MT Heilger Chemicals Pvt. Ltd.-7566 MT, Rukmani Chemicals Pvt. Ltd- 1402.590 MT and Kalina Chemicals and Rehman Chemicals - 10792 MT) as per the percentage of hazardous waste accepted by UPPCB ie. 40% (maximum) of the total Chromite Ore consumed is 23152 MT (Total hazardous waste generated by 8 industries).

- (vi) The hazardous waste storing in their premises by the above 2 industries namely Waris Chemicals & Heilger Chemicals Pvt. Ltd. is 5500 MT**
- (vii) As per the above if the total hazardous waste of 23152 MT generated and minus 5500 MT stored in the factory premises, the actual hazardous waste of 17652 MT may be considered to have been thrown in the dump.**
- (vii) As per the said letter dated 21/09/2022 of UPPCB the total hazardous waste assessed at Rania Dump, Kanpur Dehat is 85008 MT, then the balance of waste i.e. 67356 MT (85008 MT assessed - 17652 MT actual thrown) have been thrown by the other persons/industries since in the year 1976 the existence of the dump.**
- (ix) In the eye of justice of the land, it is not justified to levy the environmental compensation of the other culprits to the industries in question.**
- (x) It is also relevant that the said case as per O.A No 985/2019 and O.A. No.986/2019 is time barred and needs to be rejected by your honours.**

I, therefore, request you to please get all the information and justifications from the UPPCB before passing the order in the case.”

Today’s deliberations and further directions

13. We have considered the matter further with the help of learned Amicus, Ms. Katyayni and learned Counsel for the State PCB. The comments of the Minister of State for Women Welfare and Child Development, UP, need verification by the State PCB and Oversight Committee with reference to ground situation and remedial measures taken.

14. We find that progress in construction work of CETP at Jajmau is not satisfactory. Further, water quality of river Ganga and Kanpur Nagar which

still remains in D-category needs to be improved by preventing discharge of pollution upstream. CETP needs to remediate Chromium, Chromium need to be treated in Chromium Recovery Plant so to ensure that, chrome bearing effluents are not discharged into river Ganga or loaded on land on the name of irrigation by exploiting dilution by treated sewage effluents. The next report needs to cover the issue of existing chromium level in ground water where currently such combined effluents are used for irrigation. After installation of 20 MLD CETP, a well laid down monitorable ferti-irrigation plan will have to be laid down notifying the periods in which such effluents will be let out in Ganga and their quality. This aspect be overseen by State PCB, CPCB and NMCG.

15. Periodical health surveys be undertaken and remedial action taken in terms of safe drinking water in light thereof.

16. Further, we note that out of estimated 85,008 MT of legacy chromium waste at different locations, two companies have lifted 77,245 MT of waste and taken to the TSDF site for treatment and stabilization as per the Agreement and the work order referred to above. Work of transportation and stabilization was to be completed by 16.05.2023 i.e. today. In order to ensure that due SOPs are followed during excavation, transportation, and stabilization, UP PCB needs to monitor the process as per authorization granted.

17. With regard to remediation of ground water contamination, we do not find current status of ground water quality profile and nor of pilot project to show removal of chromium by installing RO and nano filtration system, as directed vide order dated 23.09.2022. While pump and treat method along with bio-remediation may be long term approaches, measures like RO installation at small level can quickly demonstrate

performance which may be done on priority. Possibility of stabilised waste being converted into concrete blocks may be explored.

Directions:

18. In the light of above, further status report in the matter be filed with regard to status as on 31.10.2023 by joint Committee of NMCG, State PCB, CPCB and District Magistrate, Kanpur Dehat on or before 15.11.2023 by e-mail at judicial-ngt@gov.in preferably in the form of searchable PDF/ OCR Support PDF and not in the form of Image PDF. State PCB will be the nodal agency for coordination and compliance. The report may inter alia mention:-

- i. Status of complete removal and transportation of excavated chromium waste from all the sites and waste treated and stabilized and duly ensured for stability before landfilling.
- ii. ground water profile in terms of chromium in the areas under question and installation of pilot project on chromium removal by RO/Nanofiltration.
- iii. Execution plan with timelines on ground water remediation.
- iv. Completion of CETP with respect to chromium treatment and well laid ferti-irrigation plan, existing ground water scenario and periods of discharging effluents into Ganga.

List for further consideration of 29.11.2023.

Copy of this order be forwarded to Chief Secretary, UP, Ms. Pratiba Shukla, Minister of State for Women Welfare and Child Development, NMCG, State PCB, CPCB and District Magistrate, Kanpur Dehat by e-mail for compliance.

Adarsh Kumar Goel, CP

Sudhir Agarwal, JM

Dr. A. Senthil Vel, EM

May 16, 2023
Original Application No. 985/2019
& Original Application No. 986/2019
SN